MG-1

Grand Council Peer Review — Executive Summary

Flux-modulated axial magnetic gear, three-port instrument, v0.1

Convener	James Gianotti
Program	CBJG LLC · Benchtop Motor
Rapporteur	Perplexity Computer (13th chair)
Review date	August 2026
Verdict	**Conditional pass · v0.2 required before Gate 1**

Six-seat review of the MG-1 axial magnetic gear (2/9/11, 4.5:1) and its companion artifacts MG-2 (applications), MG-3 (ferrofluid), and MG-C1 (hand-crank USB-PD charger). This summary distils the full review; companion files provide the mathematical analysis, blueprints, architectural framework, and quantum extension.

Verdict

MG-1 passes review as a bench instrument. The Atallah-Howe axial flux-modulated topology is textbook and correctly stated[1]. The three-port instrumentation closes an energy balance that leaves no unmeasured path. Ratification is **conditional** on six remedial actions being landed as GitHub issues before Gate 1.

Artifact	Verdict	Blocking conditions
MG-1 gear (v0.1)	Conditional pass	6 items (see §2)
MG-C1 charger	Conditional pass	Depends on MG-1 Gate 1 + Gate 4
MG-2 applications	Directional only	Do not spend before MG-1 Gates 1-3
MG-3 ferrofluid	Approved (QC only)	No gap flooding under any circumstance

What the artifact is

Stripped of framing, MG-1 is a coaxial magnetic gear whose ratio is $|G_r| = p_{out}/p_{in} = 9/2 = 4.5$, counter-rotating[1]. It is built into a three-port instrument that measures τ_{in}, τ_{out}, and τ_{react} simultaneously, so the static energy balance $\tau_{in} + \tau_{out} + \tau_{react} = 0$ has no unmeasured path. A hardware-latched safe-torque-off chain (SAFE-01) is independent of firmware and is the sole authority to remove torque from any rotating part[10]. RSD (Recursive State Dynamics) is the estimator layer: slip observer and thermal virtual sensor, never in the safety path. A Gate 4 falsification test cross-calibrates the torque cells and swaps them between ports, refusing any $\eta > 1$ reading as instrumentation error.

Problems the review surfaced

P-2 · Axial thrust unsized

Back-of-envelope pull-together between rotor and modulator is ~2 kN static per rotor[2]. 608ZZ bearings are marginal. **Fix: angular-contact 7000-series pair, or PTFE thrust washers.**

P-3, P-4 · Printed load path in the measurement chain

The modulator plate is printed PETG with M8 bolts in printed threads, and the reaction arm is an integral extension of the same plate. Plastic creep on the reaction arm is *direct measurement error* in the channel that closes Gate 4. **Fix: 6061-T6 aluminium modulator plate with integral aluminium reaction arm.**

P-5 · Thermal-expansion mismatch changes the air gap

Aluminium standoffs (23 µm/m/K) vs PETG frame plates (60-80 µm/m/K) over 41 mm at $\Delta T = 30$ °C changes the 0.8 mm gap by tens of µm — same order as the ferrofluid ceiling and the shim step. **Fix: publish $u_{thermal}(g)$; consider Invar standoffs.**

P-1 · No written uncertainty budget

Gate 4 is a falsification gate. Without written u(τ_{in}), u(τ_{out}), u(τ_{react}), and u(η), it is a comparison, not a falsification. **Fix: publish MG-1_uncertainty.md before Gate 3.**

P-8 · Modulator field frequency stated inconsistently

"~550 Hz" on sheet 6 vs 74.25 Hz in MG-C1 vs "hundreds of Hz" in the body. Correct ring-modulation frequency is $f = n_s \omega_{LS}/(2\pi)$ = 122 Hz at rated[4]. V1 eddy loss was over-estimated by ~20x. **Fix: single formula, one table across three speeds.**

P-9 · Reaction cell has no independent verification

Cell-swap between HS and LS ports doesn't catch a common-mode drift on the reaction cell. **Fix: add a d=5 mm hole at r = 100 mm for a dead-weight calibration before every session.**

P-10 · CBJG Customs venture wrapper is a live risk

MG-1 outputs are the highest-quality technical evidence CBJG owns; the venture thesis still contains "free-energy motor" language[11]. Without a firewall, evidence *will* be repurposed. **Fix: Council covenant — no MG data attaches to venture materials without independent calorimetry.**

Quantum computing extension

The convener's brief asks how a quantum computer can extend MG-1 theory. Three problem classes admit a defensible quantum treatment; two of them are achievable on hardware the user already accesses via the ibmq-connector project. **Quantum at MG-1 scale is a validation pipeline for MG-2, not a speed win.**

Phase	Problem class	Method	Hardware	Deliverable
Q0	2D FEA reference	Classical (femm)	Laptop	Ground-truth field map
Q1	Forward — Poisson	VQA-Poisson[6]	IBM Quantum small backend	6-qubit convergence plot
Q2	Inverse — TO	QUBO annealing[78]	D-Wave hybrid / Digital Annealer	V4 modulator pattern
Q3	Physical validation	Print V4 + Gate 1/3 measurement	Existing MG-1 rig	Measured vs predicted
Q4	Publish	Preprint + GitHub release	—	arXiv paper
Q5	Scale to MG-2	~500-cell stator TO[9]	D-Wave Advantage + FEA cluster	MG-2 stator baseline

Q3 is non-negotiable. A quantum result on MG-1 that isn't measured on the three-port instrument is a simulation, not a result. This standard matches the CBJG Customs firewall rule in P-10.

Ratification block

Result: conditional pass.

Blocking before Gate 1:

- S-1 · Aluminium modulator plate + integral reaction arm
- S-2 · Angular-contact thrust bearings or PTFE thrust washers
- S-3 · Thermal-expansion note added to drawing sheet 7
- S-4 · Modulator-frequency table added to drawing sheet 6
- S-7 · Reaction-arm dead-weight calibration hole
- S-10 · CBJG Customs firewall covenant, hoisted into every MG artifact

Blocking before Gate 3:

- S-5 · Uncertainty budget document
- S-6 · 2D FEA torque-ripple spectrum
- S-8 · Measured PMSG efficiency map
- S-9 · Pre-registered V1/V2/V3 DoE structure (3 × 4 × 3 = 36 runs, randomised)

Companion files (in this project)

MG-1_Grand_Council_Review.md — Full six-seat review — the definitive record.

MG-1_mathematical_analysis.md — Ten lemmas, sensitivity + uncertainty + eddy-loss derivations.

MG-1_blueprints_and_schematics.md — Six new drawing sheets S-16..S-21 (SVG).

MG-1_architecture_framework.md — Four-layer stack, governance, ten-rule doctrine.

MG-1_quantum_extension.md — Q0..Q5 plan and references.

MG-1_executive_summary.pdf — This document.

Sources

All URLs are clickable. Numbered footnotes throughout the body of this document map to this list.

1. Atallah, Calverley & Howe — axial-field magnetic gear (via CUHK) — https://www.phy.cuhk.edu.hk/itp/v3/links/mmm/pdfs/08R303_1.pdf

2. Torque analysis in CMG with Jiles-Atherton nonlinear FE (OSU) — https://bpb-us-w2.wpmucdn.com/u.osu.edu/dist/6/105859/files/2021/06/112_torque_analysis_in_coaxial_magnetic_gears_considering.pdf

3. Halbach axial CMG analytical model (UniWA) — https://elke.uniwa.gr/wp-content/uploads/sites/325/2024/05/%CE%94%CE%97%CE%9C%CE%9F%CE%A3%CE%99%CE%95%CE%A5%CE%A3%CE%97_%CE%A4%CE%A3%CE%9F%CE%9B%CE%91%CE%9A%CE%97%CE%A3_27-04-2024.pdf

4. Subdomain-method torque analysis, AIP Advances 13, 015018 (2023) — https://pubs.aip.org/aip/adv/article/13/1/015018/2871171/Magnetic-field-and-torque-analysis-of-coaxial

5. Politecnico di Torino thesis — CMG torque density — https://webthesis.biblio.polito.it/17238/1/tesi.pdf

6. Xu et al. — VQA for the Poisson equation, Phys. Rev. A 104 022418 — https://link.aps.org/doi/10.1103/PhysRevA.104.022418

7. Maruo et al. — TO of electromagnetic devices with Digital Annealer, IEEE TMAG — https://ieeexplore.ieee.org/document/9803271/

8. Maruo et al. — PM + magnetic-core TO (Hokkaido COMPUMAG 2021) — https://eprints.lib.hokudai.ac.jp/dspace/bitstream/2115/87018/1/COMPUMAG2021_full_maruo_31.pdf

9. Ye et al. — Quantum Topology Optimization via Quantum Annealing — https://par.nsf.gov/servlets/purl/10422510

10. SAFE-01 canonical (giomj/dev issue #14) — https://github.com/giomj/dev/issues/14

11. CBJG Customs venture verification gates (giomj/dev issue #15) — https://github.com/giomj/dev/issues/15